CS6230 Project: QR Code Decoder — Main Results

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1 Summary

This document collects the main results of the project. Every number is taken from the Design Compiler (DC), IC Compiler (ICC) and test reports in the submission folder; the full details are in **report.pdf** (microarchitecture, verification) and **pd_report.pdf** (physical design).

Best design	Netlist *r4h12* synthesised at **1.2 ns** (833.3 MHz), placed with ICC (run `pn_r4h12_L244`)
Timing	Clock period **5.0 ns → 1.2 ns** (4.17× faster); flop-to-flop path 4.98 ns → 1.19 ns; 0 violating paths in all five synthesis runs and after placement
Placement (Table 6)	**0 violating setup and hold paths**; utilisation **92.29 %** (requirement ≥ 70 %); block **243.98 × 243.45 µm**; placed cell area **52,285.97 µm²**; total power **7.3985 mW**
Functional	**120/120** cases of two full-decoder hardware sweeps, 0 mismatches; Version 40 decoded in 38,074 cycles (45.69 µs at 1.2 ns)
Verification	76 Python tests in 8 files and 9 Bluespec test benches pass with `make full`; partner benches 10/10, 5/5, V40 pass; clean rebuild from the handout passes

2 Post-synthesis results (Table 5)

Five design iterations were synthesised with Design Compiler (normal synthesis, SAED 14nm RVT, external input and output delay of 40 % of the clock period). Designs 1 and 2 are run at 5.0 ns (the plain RTL and the improved r2 RTL); designs 3–5 use the r4 RTL and tighten the clock to 2.5, 2.0 and 1.2 ns. Critical path and slack are from the f2f path group; the power is the Total row of `report_power`.

Design number	Critical path clk period (ns)	Critical path slack (ns)	Max clock freq (MHz)	Total number of leaf cells	Total number of sequential cells	Total power (mW)	Design area (µm²)
1	5.00	0.00	200.0	115,224	32,102	9.9178	216,341.52
2	5.00	0.00	200.0	90,815	36,247	4.5249	141,711.74
3	2.50	0.00	400.0	92,463	35,744	8.8798	140,112.70
4	2.00	0.00	500.0	89,407	36,657	11.9537	151,986.89
5 (best)	1.20	0.00	833.3	92,233	36,667	19.8907	143,448.02

Table 5: Post-synthesis QoR data (Design Compiler). Max clock frequency is 1000 / the period in the SDC.

From design 1 to design 5 the clock period fell by 76 % while the standard-cell area went from 60,138.2 to 55,777.5 µm² (−7.3 %). The power rises with frequency (4.52 mW at 5.0 ns to 19.89 mW at 1.2 ns) because switching activity per second grows; at equal 5.0 ns design 2 uses 54.4 % less power than design 1.

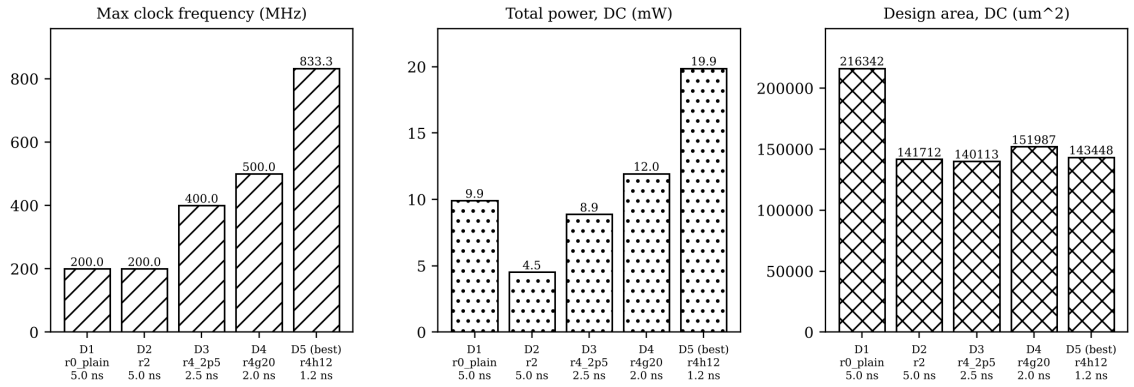


Figure 1: Frequency, power and design area for the five synthesis runs.

3 Post-placement results of the best design (Table 6)

The best design (design 5, 1.2 ns) was taken through placement with ICC (not ICC2), using the same timing libraries and SDC constraints as synthesis. Several core sizes were tried; the submitted run is the 244 µm core.

Design number	Critical path clk period (ns)	Critical path slack	Max clock freq (MHz)	Total number of leaf cells	Total number of sequential cells	Total power (mW)	Block dimensions (µm)	Design area (µm²)
1 (best)	1.20	0 violating paths (f2f path 0.80 ns)	833.3	85,490	36,667	7.3985	243.98 × 243.45	52,285.97

Table 6: Post-place QoR data (ICC, PlaceOpt). Design area is the placed standard-cell area.

- **Power:** internal 6.3528 + switching 0.9290 + leakage 0.1167 = **7.3985 mW** (from `report_power`).
- **Utilisation:** 92.29 % against the ≥ 70 % requirement; placed cell area 52,285.97 µm² in a 243.98 × 243.45 µm block.
- **Change from synthesis:** leaf cells 92,233 → 85,490 and power 19.89 → 7.40 mW after placement; the sequential cell count is unchanged at 36,667.
- **Stopping point:** the project stops at placement, so no routing was done; DRC and LVS are therefore not part of these results.

4 Core-size sweep and alternatives

The best netlist was placed with three core sizes and its neighbour netlist (1.10 ns) with two; all five runs place without violations. Tighter clocks (0.55–0.70 ns, four-bank store variant) met timing in synthesis but no placement of them was clean, so they were not selected.

Netlist	Core (µm)	Violating paths	f2f path (ns)	Cell area (µm²)	Utilisation (%)
r4h11 (1.10 ns)	242	0	0.76	52,636.24	94.41
r4h11 (1.10 ns)	244	0	0.75	52,626.12	92.89
r4h12 (1.20 ns)	242	0	0.82	52,269.68	93.75
r4h12 (1.20 ns) — submitted	244	0	0.80	52,285.97	92.29
r4h12 (1.20 ns)	246	0	0.81	52,262.04	90.57

Table 7: Place runs that were clean (all 17 PlaceOpt runs are listed in pd_report.pdf).

5 Verification results

Check	Result
Python reference-model tests (8 files)	76 tests pass
Bluespec test benches in `make test`	9 benches pass (GF, RS, syndrome, BM, blocks, corner cases, payload modes, version 40)
Full-decoder hardware sweeps (two sets)	120/120 and 120/120, 0 mismatches (30 clean, 60 at the correction limit, 30 over the limit → fail flag)
Independent cross-check with zxing-cpp	90/90 agree
Corner cases	10/10 pass (format-bit errors, tie-breaker, invalid size, uncorrectable format, …)
Partner benches against the submitted `bsv/`	10/10 corner cases, 5/5 payload modes, Version 40 Mask 3 decoded in 38,074 cycles
`make full` on the final tree	ALL VERIFICATION STEPS PASSED (10 min 7 s)
Clean rebuild from the handout with our `bsv/`	Builds and decodes the sample symbol in 563 cycles

Table 8: Verification summary. Logs are in `verification/results/`.

6 Checklist against the specification

Requirement (spec section)	Status
Module `mkQRDecoder` with the given interface, `(* synthesize *)` kept (3)	Done; `QRTypes.bsv` and `QRTables.bsv` unchanged
Functional specification G1–G4 and contract C1–C5 (4)	Done; checked by the sweeps and corner cases
Verification with one-command run and README (5)	`make test` / `make full` in `verification/`
At least four DC iterations, 40 % external delay, normal synthesis (6.2.2)	Five iterations; 40 % I/O delay in every SDC; Table 5
Placement of the best design with ICC; Table 6 (6.2.3)	Done; Table 6 above
Reports `report.pdf`, `pd_report.pdf`, `README.md`, `pd/` layout (6.3)	Present, with the requested file names